

Learning Transition-Defined Cell States from Time-Resolved Single-Cell Transcriptomics

Abstract

Time-resolved single-cell transcriptomics is used to identify the discrete populations present during development and infer how those populations give rise to one another. Many workflows define states first through expression clustering or marker-based annotation and subsequently estimate transitions between those fixed units. Recent transition-based and low-rank optimal-transport methods reduce this separation, but their latent states are not defined by an explicit criterion of predictive sufficiency. Consequently, a transport-efficient partition need not correspond to populations that are homogeneous in developmental history or potential.

We introduce a bidirectional temporal coarse-graining method for learning time-local states from a sequence of adjacent-time cell-level transport couplings. Entropic couplings are first estimated from a frozen expression representation and then held fixed. Soft state memberships are learned so that, after conditioning on the current state, individual cell identity retains little additional information about the inferred predecessor or descendant state. Under the joint distributions induced by the reference couplings and state memberships, these losses are exactly

$$I(I_t; Z_{t-1} \mid Z_t) \quad \text{and} \quad I(I_t; Z_{t+1} \mid Z_t).$$

They are optimized together with expression-coherence and transport-compression terms, while the state-level transition map is induced from the memberships rather than fitted independently.

Two degeneracies motivate the design. Fingerprints computed from a coupling that factors through the learned memberships are homogeneous within states by construction, requiring an independently estimated reference coupling. Furthermore, transition-information losses decrease when the neighboring state space is coarsened and vanish when $K = 1$. We therefore use nested model selection: entropic regularization is selected from coupling informativeness and stability, the shared state number is selected from compression, expression coherence, and stability, and transition-loss weights are selected only after both are fixed.

Because the reference couplings are derived from expression geometry, we evaluate fingerprint heterogeneity against a cross-fitted smooth geometric null and distinguish it from population-level divergence and merging. When biological replicates are available, node and edge stability is quantified by replicate-level bootstrap support. The method requires only count matrices and time labels for its core inference and returns a time-layered state-transition DAG representing transport-implied developmental compatibility rather than observed lineage.

1 Method

1.1 Overview and methodological scope

Let

$$X^{(1)}, \dots, X^{(T)}, \quad X^{(t)} \in \mathbb{N}^{n_t \times G},$$

denote count matrices collected at ordered time points

$$\tau_1 < \tau_2 < \cdots < \tau_T.$$

The method consists of two explicit stages:

1. construct cell-level optimal-transport couplings between adjacent time points;
2. jointly coarse-grain the entire sequence of fixed cell-level couplings into time-local states and a state-level transition map.

The method therefore does not claim to jointly infer the original cell-level couplings and the states. Instead, it learns a temporal coarse-graining of a fixed transport chain.

The defining principle is that a useful state should retain information about both the probable predecessors and probable descendants of its constituent cells:

time-local state = expression-compatible coarse-graining + past sufficiency + future sufficiency.

Here, sufficiency means that after conditioning on the learned state, the identity of an individual cell should contain little additional information about its predecessor- or descendant-state distribution.

1.2 Frozen expression representation

We first learn a shared expression representation

$$z_i^{(t)} = f_\psi(x_i^{(t)}) \in \mathbb{R}^d.$$

Possible choices include GLM-PCA, PCA after normalization and highly variable gene selection, or a negative-binomial latent-variable model such as scVI.

For the core method, the representation is learned before transport construction and subsequently frozen. Time point is not removed as though it were merely a batch effect.

For adjacent time points, define

$$C_{ij}^{(t)} = \frac{\|z_i^{(t)} - z_j^{(t+1)}\|_2^2}{\Delta\tau_t}, \quad \Delta\tau_t = \tau_{t+1} - \tau_t. \quad (1)$$

Alternative cost specifications are treated as sensitivity analyses rather than jointly learned components of the first implementation.

1.3 Cell-level reference couplings

Let

$$a_t \in \Delta^{n_t-1}$$

denote the cell weights at time t . Uniform weights are used when calibrated population-abundance information is unavailable.

For each adjacent pair, define

$$P_t^\varepsilon = \arg \min_{P \in \Pi(a_t, a_{t+1})} \left\{ \left\langle C^{(t)}, P \right\rangle - \varepsilon H(P) \right\}, \quad (2)$$

where

$$\Pi(a_t, a_{t+1}) = \left\{ P \geq 0 : P\mathbf{1} = a_t, P^\top \mathbf{1} = a_{t+1} \right\},$$

and

$$H(P) = - \sum_{i,j} P_{ij} \log P_{ij}.$$

After selecting ε , define the frozen reference coupling as

$$P_t^{\text{ref}} = P_t^{\varepsilon*}.$$

The matrices P_t^{ref} remain fixed throughout state learning. This is essential: if fingerprints were calculated from a coupling that factored entirely through the learned state assignments, their within-state homogeneity would hold by construction.

1.4 Informative entropy regime

The entropy parameter ε controls the amount of information retained by the reference coupling.

As

$$\varepsilon \rightarrow \infty,$$

the coupling approaches the independent coupling

$$P_t^\varepsilon \longrightarrow a_t a_{t+1}^\top.$$

Consequently, all cells acquire approximately identical transition distributions.

At very small ε , the coupling becomes sharp and may become sensitive to sampling noise and small perturbations of the cost.

We quantify coupling informativeness using

$$\mathcal{I}_t^{\text{cell}}(\varepsilon) = D_{\text{KL}} \left(P_t^\varepsilon \parallel a_t a_{t+1}^\top \right), \quad (3)$$

where

$$D_{\text{KL}}(P \parallel Q) = \sum_{i,j: P_{ij} > 0} P_{ij} \log \frac{P_{ij}}{Q_{ij}}.$$

Equation (3) is the mutual information between source- and target-cell identities under the coupling. It satisfies

$$\mathcal{I}_t^{\text{cell}}(\varepsilon) \rightarrow 0 \quad \text{as} \quad \varepsilon \rightarrow \infty.$$

A normalized version is

$$\bar{\mathcal{I}}_t^{\text{cell}}(\varepsilon) = \frac{\mathcal{I}_t^{\text{cell}}(\varepsilon)}{\min\{H(a_t), H(a_{t+1})\}}, \quad (4)$$

where

$$H(a_t) = - \sum_i a_{ti} \log a_{ti}.$$

The operating value $\varepsilon_\star$ is selected before state learning using:

1. nonzero coupling informativeness;
2. stability under biological-replicate resampling;
3. stability under small cost perturbations;
4. avoidance of nearly independent couplings;
5. avoidance of excessively sharp, unstable couplings.

No single value is treated as uniquely correct. Results are reported across an informative interval of ε .

1.5 Sparse support and feasibility

For large datasets, P_t^{ref} may be restricted to an adaptive bipartite support

$$\mathcal{S}_t \subseteq \{1, \dots, n_t\} \times \{1, \dots, n_{t+1}\}.$$

The support should include both source-to-target and target-to-source neighbors. A row-wise k -nearest-neighbor graph alone does not guarantee that a balanced coupling with the required column marginals exists.

The neighborhood size is increased until transport feasibility is achieved. Sensitivity to the support size is reported together with sensitivity to ε .

1.6 Time-local state assignments

For the core model, use a common maximum state number

$$K_t = K \quad \text{for all } t.$$

This avoids a combinatorial search over

$$(K_1, \dots, K_T).$$

The state identities remain time-local: state k at time t is not assumed to be identical to state k at another time point.

Let

$$M_t \in (0, 1)^{n_t \times K}$$

be the soft state-membership matrix, satisfying

$$M_t \mathbf{1}_K = \mathbf{1}_{n_t}. \quad (5)$$

To maintain strict positivity, parameterize

$$M_{t,ik} = \frac{\exp(U_{t,ik})}{\sum_{r=1}^K \exp(U_{t,ir})}. \quad (6)$$

Define the joint cell-state distribution

$$Q_t = \text{diag}(a_t) M_t, \quad (7)$$

and the state-mass vector

$$g_t = Q_t^\top \mathbf{1} = M_t^\top a_t. \quad (8)$$

A state is considered active for visualization only when

$$g_{t,k} > g_{\min},$$

where $g_{\min}$ is fixed before model fitting and examined by sensitivity analysis. This threshold does not alter the optimization.

1.7 Induced state-level transition map

Given the frozen reference coupling and the current memberships, define

$$T_t = M_t^\top P_t^{\text{ref}} M_{t+1} \in \mathbb{R}_+^{K \times K}. \quad (9)$$

The state-level transition matrix is therefore induced rather than freely optimized. It satisfies

$$T_t \mathbf{1} = g_t, \quad T_t^\top \mathbf{1} = g_{t+1}.$$

The corresponding conditional forward transition matrix is

$$A_t = \text{diag}(g_t)^{-1} T_t, \quad (10)$$

and the reverse conditional transition matrix is

$$B_t = \text{diag}(g_{t+1})^{-1} T_t^\top. \quad (11)$$

Under the inferred coarse-graining,

$$A_t(k, l) = \Pr(Z_{t+1} = l \mid Z_t = k),$$

whereas

$$B_t(l, k) = \Pr(Z_t = k \mid Z_{t+1} = l).$$

1.8 Low-rank reconstruction

The state assignments and induced transition matrix define a low-rank cell-level coupling

$$\hat{P}_t = Q_t \text{diag}(g_t)^{-1} T_t \text{diag}(g_{t+1})^{-1} Q_{t+1}^\top. \quad (12)$$

The reference coupling retains cell-level transport heterogeneity, while $\hat{P}_t$ is its state-level reconstruction.

Define

$$\mathcal{L}_{\text{compress}} = \sum_{t=1}^{T-1} D_{\text{KL}} \left(P_t^{\text{ref}} \parallel \hat{P}_t \right). \quad (13)$$

Because both P_t^{ref} and $\hat{P}_t$ have total mass one,

$$D_{\text{KL}} \left(P_t^{\text{ref}} \parallel \hat{P}_t \right) = \sum_{(i,j) \in \text{supp}(P_t^{\text{ref}})} P_{t,ij}^{\text{ref}} \log \frac{P_{t,ij}^{\text{ref}}}{\hat{P}_{t,ij}}. \quad (14)$$

Thus, the loss only needs to be evaluated on the sparse support of P_t^{ref} .

Strictly positive softmax memberships imply

$$\hat{P}_{t,ij} > 0$$

mathematically. In numerical calculations, use

$$\log \left(\hat{P}_{t,ij} + \delta \right)$$

with a small fixed numerical floor $\delta > 0$.

1.9 Cell-level transition fingerprints

For $t < T$, define the outgoing fingerprint

$$f_{ti}^+ = \frac{P_t^{\text{ref}}(i, \cdot) M_{t+1}}{P_t^{\text{ref}}(i, \cdot) \mathbf{1}}. \quad (15)$$

The vector

$$f_{ti}^+ \in \Delta^{K-1}$$

is the OT-implied distribution of descendant states for cell i .

For $t > 1$, define the incoming fingerprint

$$f_{ti}^- = \frac{P_{t-1}^{\text{ref}}(\cdot, i)^\top M_{t-1}}{P_{t-1}^{\text{ref}}(\cdot, i)^\top \mathbf{1}}. \quad (16)$$

The vector

$$f_{ti}^- \in \Delta^{K-1}$$

is the OT-implied distribution of predecessor states for cell i .

These fingerprints are not observed lineage distributions. They are probabilistic transition compatibilities determined by the expression representation, transport cost, entropy parameter, and neighboring state assignments.

1.10 State-level fingerprint prototypes

Define the outgoing fingerprint prototype

$$\phi_{tk}^+ = \frac{\sum_i a_{ti} M_{t,ik} f_{ti}^+}{g_{t,k}}, \quad t < T, \quad (17)$$

and the incoming prototype

$$\phi_{tk}^- = \frac{\sum_i a_{ti} M_{t,ik} f_{ti}^-}{g_{t,k}}, \quad t > 1. \quad (18)$$

These are the KL-barycenter prototypes of the cell-level fingerprints assigned to state k .

1.11 Bidirectional information-loss objective

Define the outgoing fingerprint loss

$$\mathcal{L}_+ = \sum_{t=1}^{T-1} \sum_{i=1}^{n_t} \sum_{k=1}^K a_{ti} M_{t,ik} D_{\text{KL}}(f_{ti}^+ \| \phi_{tk}^+), \quad (19)$$

and the incoming fingerprint loss

$$\mathcal{L}_- = \sum_{t=2}^T \sum_{i=1}^{n_t} \sum_{k=1}^K a_{ti} M_{t,ik} D_{\text{KL}}(f_{ti}^- \| \phi_{tk}^-). \quad (20)$$

These are soft losses. The method does not impose exact equality of all fingerprints assigned to one state.

1.12 Conditional-mutual-information interpretation

For the outgoing direction, define the joint distribution

$$p_t^+(i, k, l) = a_{ti} M_{t,ik} f_{ti,l}^+. \quad (21)$$

Under this distribution,

$$p_t^+(l \mid i, k) = f_{ti,l}^+,$$

and

$$p_t^+(l \mid k) = \phi_{tk,l}^+.$$

It follows that

$$\mathcal{L}_{+,t} = I(I_t; Z_{t+1} \mid Z_t), \quad (22)$$

where I_t denotes the identity of the observed cell at time t .

Similarly,

$$\mathcal{L}_{-,t} = I(I_t; Z_{t-1} \mid Z_t). \quad (23)$$

Therefore:

- $\mathcal{L}_{+,t}$ measures how much information about the descendant state remains in individual cell identity after conditioning on the current state;
- $\mathcal{L}_{-,t}$ measures how much information about the predecessor state remains in individual cell identity after conditioning on the current state.

A transition-sufficient state has small values in both directions.

1.13 Expression coherence

Let

$$\mu_{tk} = \frac{\sum_i a_{ti} M_{t,ik} z_i^{(t)}}{g_{t,k}} \quad (24)$$

denote the expression prototype of state k at time t .

Define

$$\mathcal{L}_{\text{expression}} = \sum_{t=1}^T \sum_{i=1}^{n_t} \sum_{k=1}^K a_{ti} M_{t,ik} \left\| z_i^{(t)} - \mu_{tk} \right\|_2^2. \quad (25)$$

This term prevents transcriptionally incompatible cells from being grouped solely because of similarities in the inferred transition structure.

1.14 Core optimization problem

For fixed $\varepsilon_\star$ and fixed K , optimize

$$\begin{aligned} \min_{\{U_t\}} \quad \mathcal{L} = & \lambda_{\text{compress}} \mathcal{L}_{\text{compress}} + \lambda_x \mathcal{L}_{\text{expression}} \\ & + \lambda_- \mathcal{L}_- + \lambda_+ \mathcal{L}_+, \end{aligned} \quad (26)$$

where the membership matrices are obtained from U_t through Equation (6).

No ℓ_1 penalty is placed on Q_t or T_t , because their total mass is fixed and their ℓ_1 norms are therefore constant.

The weights have the following interpretations:

- $\lambda_{\text{compress}}$ controls preservation of cell-level transport structure;
- λ_x controls expression coherence;
- λ_- controls predecessor sufficiency;
- λ_+ controls descendant sufficiency.

The parameters λ_- and λ_+ are selected only after $\varepsilon_\star$ and K have been fixed.

1.15 Coarsening pressure and model-selection separation

The fingerprint losses are not directly comparable across different neighboring state numbers.

In particular, if $K = 1$, then

$$f_{ti}^+ = [1]$$

for every cell, so

$$\mathcal{L}_+ = 0.$$

More generally, coarsening M_{t+1} removes information from Z_{t+1} and can reduce

$$I(I_t; Z_{t+1} \mid Z_t).$$

Therefore, the total objective in Equation (26) is not used to jointly select K and $\lambda_{\pm}$.

Instead, model selection follows the nested procedure:

1. select an informative interval of ε ;
2. select a shared K using compression, expression coherence, and stability;
3. fix ε and K ;
4. sweep λ_- and λ_+ ;
5. report sensitivity across the resulting state definitions.

The fingerprint losses may be normalized for reporting as

$$R_{+,t} = \frac{I(I_t; Z_{t+1} \mid Z_t)}{I(I_t; Z_{t+1}) + \delta_I}, \quad (27)$$

and

$$R_{-,t} = \frac{I(I_t; Z_{t-1} \mid Z_t)}{I(I_t; Z_{t-1}) + \delta_I}, \quad (28)$$

where $\delta_I > 0$ is a small numerical constant.

These ratios quantify the fraction of transition information not captured by the current state, but they are not used alone for state-number selection.

1.16 Within-state transition heterogeneity

For each state, define

$$V_{tk}^+ = \frac{1}{g_{t,k}} \sum_i a_{ti} M_{t,ik} D_{\text{KL}}(f_{ti}^+ \parallel \phi_{tk}^+), \quad (29)$$

and

$$V_{tk}^- = \frac{1}{g_{t,k}} \sum_i a_{ti} M_{t,ik} D_{\text{KL}}(f_{ti}^- \parallel \phi_{tk}^-). \quad (30)$$

These quantities are retained as diagnostics rather than being required to equal zero.

Low V_{tk}^- , low V_{tk}^+ . The state is approximately sufficient for both predecessor and descendant distributions.

Low V_{tk}^- , high V_{tk}^+ . The state contains heterogeneous outgoing transition compatibility.

High V_{tk}^- , low V_{tk}^+ . The state contains heterogeneous incoming transition compatibility.

High V_{tk}^- , high V_{tk}^+ . The state may be too coarse, or the reference transport may be unstable.

High fingerprint dispersion alone is not interpreted as biological fate priming or lineage memory.

1.17 Geometry-aware null diagnostics

Because P_t^{ref} is constructed from distances in the latent expression space, the fingerprints are themselves functions of expression geometry.

A state crossing a geometric boundary may therefore have high V_{tk}^+ or V_{tk}^- even when no distinct biological fate program exists.

To quantify geometric explainability, fit cross-fitted smooth predictors

$$\hat{f}_t^+(z) \quad \text{and} \quad \hat{f}_t^-(z)$$

using kernel regression, local regression, or a constrained neural network.

The predictor for each cell is trained without using that cell or its biological replicate.

Define

$$G_{tk}^+ = \frac{1}{g_{t,k}} \sum_i a_{ti} M_{t,ik} D_{\text{KL}} \left(f_{ti}^+ \parallel \hat{f}_t^+(z_i^{(t)}) \right), \quad (31)$$

and

$$G_{tk}^- = \frac{1}{g_{t,k}} \sum_i a_{ti} M_{t,ik} D_{\text{KL}} \left(f_{ti}^- \parallel \hat{f}_t^-(z_i^{(t)}) \right). \quad (32)$$

The interpretation is:

- high $V^\pm$ with low $G^\pm$: fingerprint heterogeneity is largely predictable from smooth expression geometry;
- high $V^\pm$ with high $G^\pm$: the heterogeneity is not captured by the specified smooth null model.

The latter is called *excess non-smooth transition heterogeneity*. It is not sufficient evidence for biological fate priming, because it may also reflect coupling noise, representation error, or an inadequate null model.

Where possible, the geometry-aware diagnostic is supplemented by:

1. local permutation tests within expression neighborhoods;
2. parametric bootstrap under a smooth transition model;
3. validation using genes not used to construct the transport cost;
4. independent lineage, velocity, perturbation, or spatial information.

1.18 Branching and merging

Define the effective number of children of state k as

$$N_{\text{child}}(t, k) = \exp \{H(A_t(k, \cdot))\}, \quad (33)$$

and the effective number of parents of state l as

$$N_{\text{parent}}(t+1, l) = \exp \{H(B_t(l, \cdot))\}. \quad (34)$$

This distinguishes population-level branching or merging from within-state heterogeneity:

Coherent divergence. $N_{\text{child}}(t, k) > 1$ while V_{tk}^+ is small. Cells in the state share a common multimodal descendant distribution.

Outgoing heterogeneity. V_{tk}^+ is large. Different cells in the state have different OT-implied descendant distributions.

Coherent merging. $N_{\text{parent}}(t+1, l) > 1$ while $V_{t+1,l}^-$ is small.

Incoming heterogeneity. $V_{t+1,l}^-$ is large. Cells in the destination state have different OT-implied predecessor distributions.

These event labels describe the inferred population transport and do not constitute observed lineage relationships.

1.19 Optimization

The induced transitions T_t , expression prototypes μ_{tk} , and fingerprint prototypes $\phi_{tk}^\pm$ are deterministic functions of the current membership matrices.

They are substituted into the objective, producing the reduced differentiable objective

$$\mathcal{L} = \mathcal{L}(U_1, \dots, U_T).$$

All dependencies of

$$T_t, \quad \hat{P}_t, \quad f_{ti}^\pm, \quad \phi_{tk}^\pm, \quad \mu_{tk}$$

on $U_1, \dots, U_T$ are included in gradient calculations.

The preferred optimizer uses full-gradient updates with backtracking line search. A proposed update is accepted only when the complete objective decreases.

Algorithm 1 Bidirectional temporal coarse-graining

- 1: Learn and freeze $z_i^{(t)} = f_\psi(x_i^{(t)})$.
 - 2: Compute adjacent-time costs $C^{(t)}$.
 - 3: Fit P_t^ε over a grid of ε .
 - 4: Select an informative interval using $\mathcal{I}_t^{\text{cell}}(\varepsilon)$ and stability.
 - 5: Fix $\varepsilon_\star$ and set $P_t^{\text{ref}} = P_t^{\varepsilon_\star}$.
 - 6: Select a shared K using compression, expression coherence, and stability, without using $\mathcal{L}_\pm$ for model selection.
 - 7: Initialize logits U_t using fine over-clustering and multiple random initializations.
 - 8: **repeat**
 - 9: Compute $M_t = \text{softmax}(U_t)$.
 - 10: Compute g_t , T_t , and $\hat{P}_t$.
 - 11: Compute f_{ti}^- , f_{ti}^+ , ϕ_{tk}^- , ϕ_{tk}^+ , and μ_{tk} .
 - 12: Evaluate the complete objective and its gradient with respect to all U_t .
 - 13: Propose a full-gradient update.
 - 14: Use backtracking line search until the complete objective decreases.
 - 15: **until** the relative objective change and membership change are below fixed tolerances
 - 16: Repeat from multiple initializations and retain a consensus solution.
 - 17: Fit geometry-aware null predictors using cross-fitting.
 - 18: Compute $V_{tk}^\pm$, $G_{tk}^\pm$, branching, and merging diagnostics.
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For datasets too large for full simultaneous optimization, a forward-backward block-coordinate procedure may be used. During each update, neighboring memberships are held fixed, and the update is damped:

$$M_t^{\text{new}} \leftarrow (1 - \gamma)M_t^{\text{old}} + \gamma\widetilde{M}_t, \quad 0 < \gamma \leq 1. \quad (35)$$

A block update is accepted only when the complete objective decreases.

Objective traces, convergence tolerances, and sensitivity to initialization are reported for every experiment.

1.20 State-number selection

A shared state number

$$K_t = K$$

is used in the core implementation.

Candidate values of K are evaluated using:

1. held-out or cross-fitted compression loss;
2. expression coherence;
3. stability across biological replicates;
4. stability across random initializations;

5. minimum active state mass;
6. synthetic state- and transition-recovery accuracy.

The fingerprint losses are not included in the criterion used to select K , because they systematically favor coarser neighboring state spaces.

After selecting K , the method reports sensitivity over

$$(\lambda_x, \lambda_-, \lambda_+)$$

at fixed K .

Time-varying K_t , split-merge proposals, and proper Bayesian state-number selection are deferred to later extensions.

1.21 Uncertainty and stability

When biological replicates are available, the complete procedure is repeated after resampling embryos, donors, cultures, or experimental replicates.

For an aligned state-level edge $k \rightarrow l$, define bootstrap support

$$S_{tkl} = \frac{1}{R} \sum_{r=1}^R \mathbb{I} \left[A_{tkl}^{(r)} > \delta_A \right], \quad (36)$$

where R is the number of bootstrap fits.

Edge width represents estimated transition mass, while edge opacity represents bootstrap support.

Resampling individual cells without resampling biological units is interpreted as algorithmic stability rather than biological uncertainty.

The following sources of sensitivity are reported separately:

1. biological-replicate resampling;
2. random initialization;
3. entropy parameter ε ;
4. latent representation and transport cost;
5. state number K ;
6. transition weights λ_- and λ_+ .

1.22 Optional global-family postprocessing

Each graph node remains a time-local state $S_{t,k}$. An optional soft assignment

$$R_t \in [0, 1]^{K \times H}, \quad R_t \mathbf{1}_H = \mathbf{1}_K,$$

maps local states to recurring global families.

Because local transition fingerprints at different times use different local state-coordinate systems, they are first mapped into global-family coordinates:

$$\tilde{\phi}_{tk}^+ = \phi_{tk}^+ R_{t+1}, \quad (37)$$

and

$$\tilde{\phi}_{tk}^- = \phi_{tk}^- R_{t-1}. \quad (38)$$

Global-family assignments may then use both expression prototypes and family-coordinate transition fingerprints.

Global families are initially fitted after local-state learning. Joint refinement of the global families and local states is not part of the core method.

1.23 Interpretive limits

The method learns a temporal coarse-graining of OT-implied transition compatibilities. It does not recover observed cell lineage.

In particular:

1. P_t^{ref} depends on the selected representation, cost, entropy, and marginal assumptions;
2. cells with identical measured profiles cannot be assigned distinct futures without additional informative structure;
3. high fingerprint dispersion may reflect expression geometry or coupling noise rather than biological priming;
4. inferred merging is population-level convergence, not proof that individual cells from distinct lineages physically merged;
5. uncalibrated cell counts are not reliable population-abundance measurements;
6. transient states occurring entirely between sampled times cannot be recovered;
7. if time and batch are fully confounded, developmental change cannot be separated from batch effects without additional assumptions.

Accordingly, the core methodological claim is:

We construct a fixed sequence of adjacent-time cell-level transport couplings and jointly coarse-grain the full sequence into time-local states that minimize the predecessor and descendant information remaining in individual cell identity after conditioning on the learned state.